

Beyond technology: Understanding digital healthcare through patient experience and sociotechnical resilience in Northern Ghana

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ABSTRACT

Purpose: Digital transformation in healthcare systems within low- and middle-income countries offers significant potential to enhance operational efficiency and improve patient care. Ghana's implementation of the Lightwave Health Information Management System (LHIMS) and its transition to the Ghana Health Information Management System (GHIMS) represent important milestones in this process. However, limited empirical research has examined patients' lived experiences during such transitions, particularly how infrastructural instability and system changes shape care across different levels of healthcare delivery. This study addresses this gap by exploring patient experiences during the LHIMS–GHIMS transition in Northern Ghana.

Methods: The study adopted a qualitative phenomenological design grounded in Sociotechnical Systems Theory (STS). Semi-structured interviews were conducted with 30 purposively selected patients across metropolitan, municipal, and rural healthcare facilities. Data were analyzed thematically to identify patterns in patient experiences with digital health system use and disruption.

Results: Two interrelated themes emerged. First, participants described digital systems as improving organization, reducing waiting times, and minimizing loss of medical records, particularly in metropolitan and municipal settings. Second, participants reported that system disruptions, such as power outages and network failures, frequently interrupted care processes, with these challenges being more pronounced in rural facilities. These findings highlight that while digitalization enhances efficiency, its benefits are conditional on infrastructural reliability and system stability.

Conclusion: Digital health transformation in resource-constrained environments is not merely a technological upgrade, but a sociotechnical process shaped by the alignment of infrastructure, organizational systems, and human actors. Sustaining its benefits requires strengthening infrastructure reliability, supporting healthcare workers, and continuously assessing patient experiences across diverse contexts to inform resilient, patient-centered digital health implementation.

1. Introduction

Digital technologies are rapidly transforming healthcare systems globally, reshaping how information is collected, managed, and exchanged [1–3]. Tools such as electronic health records (EHRs), patient portals, teleconsultations, and mobile health applications are widely recognized for their potential to enhance efficiency, reduce medical errors, facilitate continuity of care, and improve patient access, particularly in contexts constrained by workforce shortages and geographic limitations [4–6]. Across both high-income and low- and middle-income countries (LMICs), digital health initiatives are increasingly integrated into facility workflows to streamline registration, billing, clinical

documentation, and follow-up care [7–9].

In Ghana, digital health has become a central component of national health system modernization. Government and donor-supported initiatives have introduced electronic health information systems aimed at standardizing clinical processes, improving data quality, and supporting coordinated care across regions [10,11]. The Lightwave Health Information Management System (LHIMS) was one such platform implemented across public hospitals to support electronic registration, clinical documentation, and administrative processes. While prior studies have highlighted benefits such as improved record retrievability and workflow organization [10,12], these implementations have also exposed persistent challenges related to infrastructure, technical capacity, and

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contextual fit [13–15].

However, evidence increasingly shows that the presence of digital technologies alone does not guarantee improved healthcare outcomes [16,17]. The effectiveness of digital systems depends on how they are embedded within everyday clinical practice and how users experience and interact with them. Infrastructural conditions, organizational processes, and user support systems play a critical role in shaping how digital health services are experienced and sustained. These insights align with Sociotechnical Systems Theory (STS), which emphasizes that healthcare technologies operate within interconnected systems where technical components, human actors, and institutional contexts jointly influence outcomes [18,19].

Despite the growing body of research on digital health implementation in Ghana and comparable LMICs, existing studies have largely focused on system performance metrics, provider adoption, and technical feasibility. As a result, there remains a critical gap in understanding how patients experience digitally mediated care in everyday clinical settings, particularly at key service points such as registration, billing, records access, and follow-up. Where patient perspectives have been examined, they are often captured through quantitative measures of satisfaction or acceptance, which do not fully reflect the nuanced and context-dependent nature of patient experiences [20].

This gap is especially pronounced in Northern Ghana, a setting characterized by significant infrastructural variability, digital inequities, and sociocultural diversity that may shape digital health experiences differently from better-resourced urban environments [5,7]. Furthermore, although the transition from the Lightwave Health Information Management System (LHIMS) to the Ghana Health Information Management System (GHIMS) represents an important policy and system change, there is limited empirical evidence on how such transitions affect patients' experiences, particularly in terms of trust, continuity of care, and perceived system reliability [10,13,21]. The absence of such insights has important implications, as patient experience is increasingly recognized as central to service quality, sustained use, and equity in digital health systems [20,22–26].

In sum, while prior research has advanced understanding of digital health implementation from technical and provider perspectives, there remains a need for context-sensitive, qualitative inquiry into patients' lived experiences, particularly in environments where infrastructural conditions and service delivery processes are variable and interdependent.

To address this gap, the present study explores how patients in Northern Ghana experience digital health services during routine clinical encounters, with attention to key points of care including registration, records, billing, and follow-up. Grounded in Sociotechnical Systems Theory (STS), the study conceptualizes patient experience as an outcome of the interaction between technological systems, infrastructural conditions, and organizational processes. Specifically, the study addresses two research questions: 1) How do patients in Northern Ghana describe their experiences with digital health services across key points of care? 2) What barriers and enabling factors shape these experiences, particularly in the context of system disruptions and transitions (e.g., the LHIMS–GHIMS migration)?

These questions are significant for two main reasons. First, they extend Ghanaian digital health research by centering patient experiences, providing insights that complement existing provider- and system-focused studies [12,21]. Second, by examining experiences within a context of infrastructural variability and system change, the study generates evidence to inform efforts aimed at improving the reliability, responsiveness, and equity of digital health systems in resource-constrained settings [22,25].

To achieve this aim, the study adopts a qualitative phenomenological approach, which is well suited to capturing the depth and complexity of lived experiences and how individuals interpret digitally mediated healthcare environments. By foregrounding patients' voices, the study contributes context-specific evidence to support the development of

more resilient, equitable, and patient-centered digital health systems in Ghana.

2. Theoretical framework

This study is guided by Sociotechnical Systems Theory (STS), which provides an appropriate framework for examining patients' experiences of digital health services in Northern Ghana. STS, originally developed by Trist and Bamforth [27] and later extended by Emery and Trist [28], emphasizes that the outcomes of any technological system are produced through the interaction between technical components and social arrangements. In healthcare settings, this means that the functioning and consequences of digital systems cannot be understood solely in terms of software or hardware, but must also be examined in relation to organizational workflows, human actors, institutional routines, and the broader infrastructural environment.

The relevance of Sociotechnical Systems Theory (STS) to the present study lies in the fact that patients do not encounter digital health systems as isolated technologies; rather, they experience them through a network of interdependent sociotechnical processes involving registration staff, nurses, records officers, cashiers, clinicians, electricity supply, internet connectivity, and institutional protocols. Patients' experiences of waiting time, service efficiency, continuity of care, trust, and frustration are therefore shaped not only by the existence of digital systems, but by how effectively the technical and social components of care delivery function together [27,29].

In the context of Northern Ghana, the implementation and use of digital health systems are embedded within conditions of uneven infrastructure, intermittent electricity, unstable network connectivity, staff workload, and varying institutional capacities across urban, municipal, and rural health facilities. STS is particularly suitable for analyzing such contexts because it enables patient experiences to be interpreted as outcomes of alignment or misalignment between technical systems and social systems [30,31]. When digital infrastructure functions reliably and staff can integrate it effectively into care delivery, patients may experience greater efficiency, improved coordination, and stronger continuity of care. Conversely, when there are infrastructural failures, workflow breakdowns, or inadequate technical support, patients may experience delays, confusion, and diminished confidence in the care process [32,33].

Thus, STS provides the central analytical lens for this study by framing patient experiences as sociotechnical outcomes rather than as individual reactions to technology alone. It allows the study to move beyond a purely technological interpretation of digital health and instead examine how healthcare delivery is co-produced through the interaction of systems, people, institutions, and infrastructure [34,35]. This makes STS especially appropriate for understanding digital healthcare in resource-constrained settings such as Northern Ghana, where the effectiveness of technological innovation depends heavily on the resilience of the surrounding social and organizational environment.

3. Conceptual frameworks

The conceptual framework for this study is derived from Sociotechnical Systems Theory (STS). It conceptualizes patients' experiences of digital health services as outcomes of the interaction between technical conditions, social processes, and organizational arrangements within healthcare delivery.

At the technical level, the framework considers factors such as electricity supply, internet connectivity, system functionality, and the digital platforms used in registration, billing, records management, and follow-up care. At the social and organizational level, it considers staff support, workflow coordination, institutional routines, and the ways health workers mediate patients' encounters with digital systems during service delivery.

Within this framework, patient experiences are not treated as direct

responses to technology use in isolation, but as products of a digitally mediated care environment. Patients experience the consequences of sociotechnical alignment through outcomes such as service efficiency, reduced confusion, continuity of care, trust in service delivery, and satisfaction with hospital processes. Conversely, when the technical and social elements are poorly aligned, patients may experience delays, frustration, discontinuity, and uncertainty.

The framework, therefore, positions patient experience as an emergent outcome of sociotechnical interaction, making STS an appropriate lens for understanding digital healthcare delivery across different levels of healthcare infrastructure in Northern Ghana. As shown in Fig. 1. (See Tables 1 and 2.)

4. Materials and methods

4.1. Philosophical orientation

This study is grounded in transcendental phenomenology, which seeks to understand and describe individuals' lived experiences as consciously perceived, free from prior assumptions [36,37]. This philosophical orientation is appropriate for examining how patients experience and make meaning of digitally mediated healthcare interactions within their everyday clinical encounters.

Central to this approach is the practice of epoché (bracketing), whereby the researchers consciously set aside preconceived notions and prior knowledge about digital health systems to remain open to participants' perspectives [36,37]. This was achieved through continuous reflexive awareness during data collection and analysis, ensuring that interpretations were grounded in participants' narratives rather than researcher assumptions. The study also draws on Sociotechnical Systems (STS) theory as an interpretive lens, recognizing that patient experiences of digital health systems are shaped by the interaction between technological tools, human actors, and organizational and infrastructural contexts. This perspective complements the phenomenological focus by situating individual experiences within broader systemic conditions.

In line with phenomenological principles, bracketing (epoché) was practiced through continuous reflexive awareness, enabling the researchers to set aside prior assumptions and remain grounded in participants' lived experiences [37]. During analysis, meaning units were derived through line-by-line examination of transcripts to identify significant statements related to participants' experiences [36,37]. These were grouped into patterns through iterative comparison across participants. Essence synthesis involved integrating these patterns into themes that captured the core structure of participants' experiences, reflecting how digital health systems were perceived and navigated within the

Table 1
Theoretical Mapping and Constructs Guiding the Study.

Study Variable	STS Dimension	Role in the Study
Infrastructure reliability	Technical subsystem	Examines how electricity supply, internet connectivity, and system functionality shape patient experiences
Staff support and mediation	Social subsystem	Explores how frontline staff explain, operate, and sustain digitally mediated service delivery for patients
Organizational workflow and coordination	Joint optimization of social and technical systems	Assesses how digital systems interact with registration, billing, records, and consultation processes
Continuity of care	System integration	Interprets how digital records and coordinated processes support repeated visits and follow-up care
Experiences during system disruptions	Sociotechnical misalignment	Analyzes how breakdowns in infrastructure or systems affect waiting time, access, and patient confidence
Patient trust in digitally mediated care	Emergent sociotechnical outcome	Examines how confidence in care is shaped by the reliability and coordination of both human and technical elements
Perceived efficiency of services	Emergent sociotechnical outcome	Interprets patients' views of speed, orderliness, and reduced confusion as products of system alignment

Source: Authors' Construct (2026)

Table 2
Theme Generation tool.

1	Hospital
2	Digital
3	Time
4	Network
5	Process
6	Computers

sociotechnical context of care delivery.

4.2. Research approach

A qualitative research approach was employed to explore patients' lived experiences with digital health systems in Northern Ghana. This

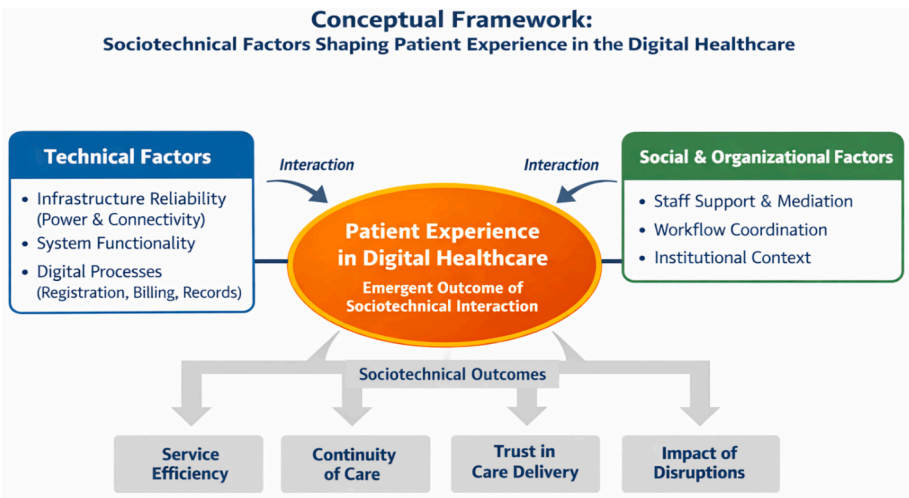


Fig. 1. Conceptual framework.

approach emphasizes depth, context, and meaning-making, making it well-suited to examining how patients perceive access, usability, and trust in digitally mediated care across diverse sociocultural and infrastructural environments [36].

The study recognizes that patient experiences are embedded within broader sociotechnical conditions, including staff mediation, institutional responsiveness, and infrastructural reliability [19,38]. Qualitative inquiry was therefore used to address “how” and “why” questions regarding patients' interactions with digital health services in real-world settings [39].

4.3. Study design

This study employed a qualitative phenomenological design to explore patients' lived experiences of digital health systems across different levels of healthcare delivery in Northern Ghana. The phenomenological approach was appropriate for capturing how individuals perceive, interpret, and make meaning of their interactions with digitally mediated healthcare services within real-world clinical contexts [36,37]. The study was situated within a Sociotechnical Systems (STS) perspective, which conceptualizes healthcare delivery as an interaction between technological systems, human actors, and organizational and infrastructural conditions. This orientation enabled the study to examine patient experiences not only as individual perceptions but as outcomes shaped by the alignment or misalignment of these interconnected elements.

4.4. Data collection

Data were collected through semi-structured, in-depth interviews developed in line with the study objectives. The guide covered key domains, including experiences with digital registration, health records, waiting time, system efficiency, staff interaction, continuity of care, and system disruptions. The interviews were conducted in English or the local dialect, Dagbani, through face-to-face conversation. Each interview session lasted approximately 20–40 min and used open-ended questions with follow-up prompts to elicit detailed accounts of participants' lived experiences. Dagbani interviews were professionally translated and transcribed for accuracy. A bilingual researcher handled the initial translation, with an independent review and back-translation of key sections to ensure consistency with the original responses.

Data saturation, defined as the point at which no new information or themes emerge, served as the primary criterion for determining sample size adequacy [40–42]. In this study, saturation was reached after approximately 26 interviews, where recurring patterns in participants' experiences with digital health services became evident across the three study sites. To confirm this, additional interviews were conducted, resulting in a total of 30 participants, with no substantially new themes emerging.

The study focused on issue-specific saturation related to patients' experiences with digital health systems, particularly across key areas such as digital registration, health record management, waiting time, system disruptions, and staff interaction. The final sample provided sufficient depth and diversity of perspectives across metropolitan, municipal, and rural healthcare settings, ensuring a comprehensive understanding of the sociotechnical factors shaping patient experiences [43,44].

4.5. Study setting

The study was conducted in three government hospitals in Northern Ghana, purposively selected to represent different levels of healthcare delivery and digital infrastructure. These sites made it possible to compare how patients felt about their care across the rural–urban spectrum and different levels of healthcare delivery.

- Tamale Teaching Hospital (TTH) is a tertiary referral and teaching hospital located in an urban setting with relatively advanced digital infrastructure and active engagement in national system transitions, including the shift from LHIMS to GHIMS.
- Savelugu Municipal Hospital is a secondary-level facility serving peri-urban and rural populations, characterized by moderate resource constraints and intermittent digital connectivity.
- Tolon District Hospital is a district-level rural facility with limited infrastructure, reflecting the challenges of implementing digital health systems in peripheral settings.

The study sites were purposively selected to capture variation across three levels of healthcare delivery in Northern Ghana: tertiary (Tamale Teaching Hospital), secondary (Savelugu Municipal Hospital), and primary (Tolon District Hospital). The selection was guided by differences in patient volume, level of digital system implementation, and infrastructural capacity. This approach enabled the study to capture diverse patient experiences across urban, municipal, and rural healthcare contexts, rather than reflecting arbitrary site selection.

4.6. Participants

Participants were adult patients (18 years and above) who had recently accessed healthcare services through digital health systems at the selected hospitals. Eligibility required direct experience with digital service processes such as electronic registration, billing, records management, or follow-up systems.

4.7. Sampling and recruitment

A purposive sampling strategy was employed to recruit participants with direct, first-hand experience of digital health services, consistent with phenomenological research principles [36,37]. Prior to data collection, key sampling criteria were established to ensure variation in participant characteristics, including age, gender, service department (e.g., outpatient, pharmacy, laboratory, consultation, etc.), and frequency of healthcare utilization. These criteria were defined to capture a broad range of patient experiences across different points of interaction with digital health systems [45,46].

Participant recruitment was operationalized through a consecutive sampling approach within a purposive framework. Eligible patients were identified and screened by members of the research team, with support from outpatient and records department staff who facilitated access to patients meeting the inclusion criteria. Screening was conducted based on pre-defined eligibility requirements, including prior engagement with digital health services at the facility.

Participants were approached consecutively at the end of their clinical visits during the study period to minimize disruption to care. The study was explained to them, and those who expressed willingness to participate provided informed consent before inclusion. Recruitment continued iteratively at each facility until the predetermined sample size of 10 participants per hospital (total $n = 30$) was achieved.

To ensure balanced representation across departments and participant characteristics, recruitment was continuously monitored during data collection. The research team tracked the distribution of participants across key variables (e.g., department, gender, age group) and adjusted recruitment accordingly to avoid overrepresentation of any single group. This approach ensured diversity in the sample without relying on arbitrary selection.

4.8. Theme identification and data analysis

Interview data were analyzed thematically to identify recurring patterns and meanings across participants' accounts. The analysis followed an iterative process. First, all interview recordings were transcribed verbatim and read repeatedly to achieve immersion in the data.

Where interviews were conducted partly in Dagbani, translation into English was undertaken during transcription by researchers proficient in both languages. Second, transcripts were reviewed line by line to identify meaningful units relevant to the study objectives. Initial codes were then generated inductively from the data, while interpretation was guided by the broader sociotechnical orientation of the study.

The coded data were compared across participants and study sites to identify patterns of similarity and difference. Related codes were subsequently grouped into broader categories, from which themes were developed and refined through constant comparison and repeated review of the transcripts. This process enabled the identification of both shared and context-specific experiences across metropolitan, municipal, and rural facilities. The final themes were selected based on their relevance to the research questions and their ability to capture the socio-technical dimensions of patients' experiences.

Voyant Tools Software was used as a supplementary aid to support word pattern exploration and theme development. Features such as Cirrus, Links, and Trends helped visualize recurring terms and their relationships within the data. However, theme generation did not rely solely on software outputs; rather, themes were developed through interpretive engagement with participants' narratives and iterative comparison across transcripts. See Fig. 2.

4.9. Ethical considerations

All procedures conducted in this study adhered to the ethical standards established by the University for Development Studies, Ghana. Our research received clearance from the Faculty of Communication and Media Studies Ethics Review Committee (approval no 180-FCMS-ERC/2025/12). Participants were granted informed consent before enrolling in the study. All participants gave voluntary informed consent prior to engaging in the in-depth interviews. The informed consent procedure encompassed guarantees of confidentiality, anonymity, and the freedom to withdraw at any time without repercussions.

4.10. Trustworthiness

To ensure the rigor and trustworthiness of the study, several strategies were employed in line with established qualitative research criteria [36,47]. Credibility was enhanced through prolonged engagement with

the data, iterative analysis, and the use of verbatim participant quotations to ground interpretations in participants' lived experiences.

Dependability was supported through a systematic and transparent analytical process, including consistent coding procedures and iterative theme development. An audit trail was maintained to document key decisions made during data collection and analysis, ensuring transparency and methodological consistency [47].

Confirmability was strengthened through reflexive awareness, with the researchers remaining mindful of their potential influence on data interpretation and ensuring that findings were grounded in participants' narratives rather than researcher assumptions [36].

Transferability was facilitated through detailed descriptions of the study context, healthcare settings, and participant characteristics, allowing readers to assess the applicability of the findings to similar contexts [47].

Fig. 2 illustrates the terms utilized repeatedly by participants and their connections to other pertinent keywords within the data. Several pertinent keywords utilized repeatedly by individuals, as illustrated in the Cirrus, include “digital”, “computers”, and “network”. The links section demonstrates words like “registration”, “time”, “hospital”, and “process”. The more dominant words were grouped into the major themes.

Fig. 3 illustrates the constructed categories that emerged from the data analysis process. These constructed categories represent the key dimensions underlying participants' experiences and perspectives. Together, they provide a structured framework for understanding the phenomenon under investigation and serve as the foundation for the subsequent presentation of results.

Frequently used words + Constructed Categories = Major.

5. Results and analysis

Table 3 presents the distribution of study participants across the selected healthcare facilities, including facility type, location context, sample size, and the eligibility criteria used for participant selection.

The sociodemographic characteristics of participants are presented in Supplementary Table S1.

This section presents the findings organized through a Sociotechnical Systems (STS) lens, which conceptualizes patient experience as an outcome of the interaction between technical conditions (e.g.,



Fig. 2. Theme Identification.
Source: Voyant tool (2026).

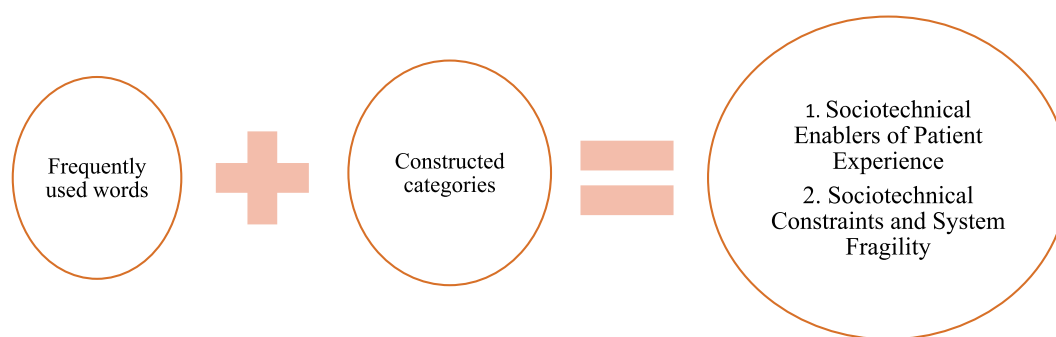


Fig. 3. Constructed Categories.

Table 3
Sample Distribution Across Study Sites.

Study Site	Facility Type	Location Context	Actual Sample Size (n)	Selection Criteria
Tamale Teaching Hospital (TTH)	Metropolitan /Tertiary Hospital	Urban-Regional referral and teaching facility	10 participants	Adult patients (≥ 18 years) who have accessed digital health services such as electronic registration, billing, records, or follow-up systems
Savelugu Municipal Hospital	Municipal /Secondary Hospital	Peri-urban Mid-level healthcare facility	10 participants	Adult patients (≥ 18 years) with recent exposure to digital health systems and processes
Tolon District Hospital	District / Primary Hospital	Rural Community-level healthcare facility	10 participants	Adult patients (≥ 18 years) who have interacted with digital processes during healthcare encounters
Total	–	–	30 participants	Sampling continued until data saturation was achieved [40]

Source: Authors' Construct (2026).

infrastructure, digital systems) and social–organizational processes (e. g., staff mediation, workflow coordination).

The analysis draws on participants' experiences across three levels of healthcare delivery: Tamale Teaching Hospital (metropolitan/tertiary), Savelugu Municipal Hospital (municipal/secondary), and Tolon District Hospital (rural/primary), highlighting both points of alignment (where systems function effectively) and points of breakdown (where socio-technical fragility disrupts care). The findings are organized under two major themes derived from participants' experience:

1. Theme 1: Sociotechnical Enablers of Patient Experience
2. Theme 2: Sociotechnical Constraints and System Fragility

5.1. Theme 1: Sociotechnical Enablers of Patient Experience

Across all study sites, participants consistently described digital systems as enhancing the organization, coordination, and flow of healthcare services. These experiences reflect moments of sociotechnical alignment, where digital tools are effectively integrated into service delivery processes.

One participant explained:

“Emmmm.... Once I enter the Tamale Teaching Hospital, the electronic registration system is the first thing that I come across. Registering till billing, it is now all done by computers, and this process is much more organized compared to the paper system.” (P01 at TTH).

Similarly, another participant noted:

“Everything at the pharmacy and cashier stations is all electronically maintained- in terms of billing, as well as receipt issue...ummm... It is a sense that the hospital has become very dependent on digital processes. I believe it will assist them in the process of monitoring patients at a better pace, and it also accelerates the whole process of work” (P03, Tamale Teaching Hospital).

The participant views above illustrate how digital health systems have been institutionalized as integral components of routine healthcare delivery, rather than functioning as isolated technological tools. Participants' narratives show that digital processes span multiple service points, registration, pharmacy, cashier services, and clinical

coordination, thereby creating a continuous and interconnected care pathway. The emphasis on organization and speed suggests that patients perceive improvements not only in efficiency but also in the predictability and structure of service delivery, which reduces uncertainty during care encounters.

From a sociotechnical perspective, these outcomes reflect a high level of alignment between the technical subsystem (digital platforms) and the organizational subsystem (hospital workflows and staff practices). The effectiveness of the system is therefore not merely a result of digitization itself, but of its successful integration into everyday clinical and administrative routines. Additionally, the perception that hospitals have become “dependent on digital processes” indicates a transition toward digitally mediated healthcare environments, where service delivery is increasingly reliant on system functionality. This dependency enhances coordination and monitoring but also implies that any disruption in the system can have far-reaching consequences. Overall, these findings indicate that the benefits of digital health systems are realized through coordinated interaction between technology, institutional processes, and human actors, reinforcing the core principles of Sociotechnical Systems Theory.

5.2. Organizational workflow and interdepartmental coordination

Participants further emphasized how digital systems facilitate interdepartmental coordination, enabling faster information flow and reducing delays in service delivery. One of the participants revealed that.

“In this hospital, the first thing that I encounter is the electronic registration. The employees key in my information into a computer system and hand me a printed slip. My information is then already available to the doctor, so I no longer have a physical folder. Outputs at the lab are digitally uploaded and a cashier at the time of payment is also computerized. Ummmm....My next appointment date is also introduced in the system, and it becomes easier to follow up with it. All that, registration to billing has become computerized and I find it more organized than the paper system in the past”. (P05 at TTH).

Similarly:

“It facilitates interdepartmental communication. As soon as my lab

results were uploaded, the doctor became aware of it. That would take days in the past before" (P19 at Saveluge Hospital).

The views illustrate how digital systems enable end-to-end integration of healthcare processes, linking registration, clinical consultation, laboratory services, billing, and follow-up into a unified workflow. The elimination of physical folders and the real-time availability of patient information enhance continuity of care and reduce delays associated with manual record handling. Participants' experiences further indicate that digital platforms improve interdepartmental communication, allowing clinical decisions to be made more efficiently. From a sociotechnical perspective, this reflects joint optimization, where technological systems and organizational routines are effectively aligned to support coordinated service delivery. The improved patient flow and reduced fragmentation highlight how digital integration restructures care processes into a more coherent and responsive system. Overall, these findings show that the value of digital health systems lies in their ability to synchronize multiple service points within a single operational framework.

5.3. Staff mediation and patient interaction

Although patients are not direct users of digital technologies, their experiences are shaped through mediated interactions with healthcare staff, who serve as critical intermediaries between the system and the patient. Patients encounter digital systems through processes such as registration, billing, and consultation, all of which are facilitated by staff. This highlights that patient experience is not determined solely by technology, but by how effectively staff interpret, operate, and communicate digital processes within care delivery. Participants widely associated digital systems with increased efficiency, reduced waiting times, and improved reliability, which contributed to enhanced trust in healthcare delivery.

"Yes, for me I see the electronic records are time-saving and prevent misunderstandings. I would lose folders some years ago- now this is no longer the case. Everything is recorded in the computer". (P08 at TTH).

"I think the digital process will allow me to complete it promptly. I no longer have to wait long long queues to go through the paper system. My history is soon documented and even the doctor is aware of my previous visit" (P16 at Saveluge Hospital).

These illustrate how digital health systems enhance both operational efficiency and patient confidence in care delivery. The reduction in lost records and waiting times reflects improved data reliability and streamlined workflows, which minimize errors and administrative delays. Participants' emphasis on timely access to medical history also indicates strengthened continuity of care, enabling more informed clinical decisions. From a sociotechnical perspective, these outcomes emerge through the interaction between reliable digital systems and effective staff facilitation, rather than technology alone. Consequently, efficiency and trust are co-produced as patients experience more organized, predictable, and dependable healthcare processes.

Theme 2: Sociotechnical Constraints and System Fragility

Despite the benefits of digitalization, participants, particularly in rural settings, emphasized that system effectiveness is highly dependent on infrastructure stability, especially electricity and internet connectivity. It was in line with this that a participant explained.

"It makes hospital work smoother when the light and network are good. The doctor can quickly check my records. I think it has made the process modern, but in our area, we only enjoy it on good days when everything works." (P30 at Tolon Hospital).

Another participant also added to the above establishment.

"The digital registration helps when there is light. It is quicker because they type my name and print the slip in minutes... Now, when the system is working, it saves time and reduces confusion at the records office." (P18 at Tolon Hospital).

These findings show that perceived efficiency is conditional, emerging only when the technical infrastructure is stable. From an STS

perspective, this reflects incomplete sociotechnical alignment, where infrastructural limitations constrain the full realization of technological benefits.

5.4. System disruptions and sociotechnical breakdown

Participants across all study sites described how system failures disrupt care delivery, exposing the fragility of digitally dependent healthcare systems. These disruptions, often triggered by power outages, network instability, or system downtime, were not experienced as isolated technical glitches but as system-wide breakdowns that affected multiple stages of care simultaneously, including registration, billing, consultation, and records management.

"At one point, the Lightwave system ceased to work at all. They informed us that it was Accra trouble, so all was returned to paper. That was a very confusing day since the nurses were again jotting everything down manually. The lines were so big, and we even wasted nearly the entire morning registering..." (P01 at TTH).

Another participant also underscored when asked about the system disruptions.

"Yes, on one occasion the network was downed to nearly two hours. All waited; nobody was able to check in or make payments..." (P03 at TTH).

In a similar vein, one of the participants explained.

"At one such follow-up, the power went out. The employees were forced to halt until the generator was running. I had to wait approximately an hour to see the consultant" (P05 at TTH).

These views above collectively indicate how disruptions in digital infrastructure trigger system-wide breakdowns in healthcare delivery, rather than isolated technical inconveniences. The failure of the Lightwave system and subsequent reliance on manual processes demonstrates how deeply embedded digital systems have become in routine workflows, such that their absence leads to confusion, delays, and loss of coordination. Similarly, network downtime and power outages reveal the extent to which core service points, registration, billing, and consultation are tightly interconnected and dependent on continuous system functionality. From a sociotechnical perspective, these experiences reflect a misalignment between the technical subsystem and organizational processes, where the breakdown of infrastructure disrupts the entire care continuum. The prolonged waiting times and halted services further highlight the high degree of dependency on centralized digital systems, which limits local flexibility in responding to failures. Overall, these disruptions expose the fragility of digitally mediated healthcare systems and underscore the need for more resilient infrastructural and operational support mechanisms.

5.5. Sociotechnical fragility in resource-constrained settings

Participants' accounts reveal that while digital health systems are perceived as modern and efficient, their performance remains highly contingent on the stability of the underlying infrastructure. In resource-constrained settings, frequent disruptions in electricity and network connectivity expose the inherent fragility of these systems, limiting their reliability and consistency in everyday healthcare delivery.

"Everything is running fine when the lights are on... However, as soon as the network is turned off, all is terminated. It is contemporary and impressive yet delicate (P19 at Saveluge Hospital).

Another participant also narrates,

"At least on each visit, there is a challenge. The lights will go off frequently, and the computers will crash..." (P22, at Saveluge Hospital).

The paradoxical nature of digital healthcare systems in resource-constrained settings, where efficiency and fragility coexist. While participants acknowledge the system as "contemporary and impressive," they simultaneously emphasize its vulnerability to frequent infrastructural disruptions. The reliance on stable electricity and network connectivity means that even minor interruptions can halt entire service

processes, as reflected in repeated experiences of system crashes and power outages. From a sociotechnical perspective, this illustrates that system performance is not determined solely by technological capability but by the stability of the surrounding environment. The recurring nature of these disruptions further normalizes instability within care delivery, shaping patient expectations and experiences. This dual perception reinforces the idea that digital systems, though transformative, remain conditionally effective, dependent on the alignment between technical infrastructure and contextual realities.

5.6. Staff adaptation and sociotechnical resilience

Despite recurring system failures and infrastructural instability, participants consistently emphasized the critical role of healthcare staff in sustaining service delivery. Their accounts reveal that staff adaptability, problem-solving, and commitment function as key stabilizing forces within digitally mediated care environments. This highlights the importance of the social subsystem in compensating for technical limitations, where human agency enables continuity of care even when digital systems are disrupted. In this way, resilience in digital healthcare emerges not solely from technological robustness, but from the capacity of healthcare workers to adapt, improvise, and maintain operational flow under constrained conditions.

“They lead us in a polite manner. I have even encountered clerks staying late to input missed information...” (P03 at TTH).

“The system is good when the IT officer is present. Rapid technical support is significant...” (P18 at Savelugu Health care).

“For me, using the digital system has been a learning experience... I don't touch the computers myself, but I interact with the staff...” (P12 at TTH).

The central role of human agency in sustaining digitally mediated healthcare systems, particularly in contexts where technical reliability is inconsistent. Participants' experiences reveal that staff do not merely operate digital systems but actively mediate, stabilize, and extend their functionality through adaptive practices. The reference to clerks working beyond normal hours to input missed data reflects a commitment to maintaining data continuity and system integrity, even after disruptions. Similarly, the emphasis on the presence of IT personnel highlights the importance of localized technical expertise in restoring system functionality and minimizing downtime. Furthermore, the account of indirect interaction with the system through staff reinforces the idea that patients experience digital healthcare primarily through human intermediaries, rather than direct technological engagement. Staff adaptability, problem-solving, and communication therefore function as key mechanisms of sociotechnical resilience, bridging the gap between technological limitations and patient needs. This indicates the effectiveness of digital health systems is not solely determined by technological robustness, but by the capacity of human actors to sustain and adapt system operations in practice.

6. Discussion

This study explored patients' experiences with digital health services across three levels of healthcare delivery in Northern Ghana, revealing that these experiences are shaped by a dynamic interaction between technological systems, infrastructural conditions, and human agency. Interpreted through the lens of Sociotechnical Systems Theory (STS), the findings demonstrate that digital healthcare is not merely a technical intervention but a socially embedded system, where outcomes depend on the degree of alignment between technical infrastructure, organizational workflows, and human actors.

The findings suggest that digital health effectiveness is conditional and context-dependent, shaped by the alignment between technological systems, infrastructure, and organizational processes. While prior studies emphasize sociotechnical alignment [18,19], this study extends the literature by showing how infrastructural instability, such as power

outages and network disruptions, actively reshapes patient experiences. In this context, perceived efficiency, continuity, and trust are not fixed attributes but emerge dynamically from local conditions. This highlights sociotechnical alignment as an ongoing, context-sensitive process and positions digital health systems as fluid and adaptive configurations, rather than static technological solutions.

The findings further show that patient experiences emerge as sociotechnical outcomes, reflecting both the strengths and vulnerabilities of digitally mediated healthcare environments. When technical systems function reliably and are effectively integrated into service delivery processes, patients experience improved efficiency, coordination, and continuity of care. Conversely, when infrastructural instability or system disruptions occur, these benefits are undermined, revealing the fragility of digital health systems in resource-constrained settings.

6.1. Sociotechnical alignment as a driver of efficiency, coordination, and trust

Across all study sites, participants described digital health systems as enhancing service efficiency, record management, and interdepartmental coordination. From an STS perspective, these outcomes reflect moments of sociotechnical alignment, where digital technologies are successfully integrated with institutional workflows and supported by staff practices.

The replacement of paper-based systems with electronic processes improved the organization of registration, billing, and clinical documentation, reducing delays and minimizing the risk of lost records. These improvements contributed not only to operational efficiency but also to patients' confidence in the reliability and professionalism of healthcare services. In this sense, digital systems function both as practical tools and as symbolic markers of modernization and institutional credibility.

These findings are consistent with studies in low- and middle-income settings, which show that digital health systems can enhance administrative efficiency, improve access to patient information, and strengthen coordination across service units [25,48,49]. However, the present study extends this understanding by demonstrating that these benefits are not inherent in the technology itself but emerge through the effective integration of technical and social components within healthcare delivery.

Importantly, the findings also highlight that efficiency is context-dependent. Participants in rural facilities emphasized that digital systems are effective only “when there is light,” underscoring the critical role of infrastructural stability. This reinforces the STS principle that technological performance is contingent upon the broader system within which it operates. In settings where electricity and connectivity are unreliable, the potential benefits of digitalization remain uneven and conditional.

6.2. Infrastructural fragility and sociotechnical misalignment

The second major finding reveals that digital health systems in Northern Ghana are characterized by structural fragility, arising from the interdependence between technical systems and infrastructural conditions. Participants' experiences of system failures, particularly during network outages and power interruptions, demonstrate how disruptions in the technical subsystem can lead to system-wide breakdowns in service delivery.

The reported breakdown of the Lightwave system at Tamale Teaching Hospital illustrates how centralized digital systems are vulnerable to infrastructural and systemic disruptions. When the system failed, healthcare delivery reverted to manual processes, resulting in confusion, long waiting times, and reduced service efficiency. From an STS perspective, this reflects a state of sociotechnical misalignment, where the technical system is not adequately supported by the infrastructural and organizational environment.

These findings align with existing literature on sociotechnical risk, which emphasizes that digital health systems in resource-constrained settings are particularly susceptible to instability when technological implementation outpaces infrastructural readiness [18,22,38]. The study further shows that patients are not passive recipients of these disruptions; rather, they develop an awareness of the system's dependence on infrastructure, recognizing that failures are not isolated incidents but part of a broader systemic vulnerability.

6.3. Human agency and sociotechnical resilience

Despite these challenges, the findings underscore the critical role of human agency in sustaining healthcare delivery. Participants consistently highlighted the efforts of healthcare staff in managing system disruptions, maintaining communication, and ensuring continuity of care.

From an STS perspective, this reflects the importance of the social subsystem in stabilizing technological systems. Staff members act as mediators who bridge the gap between technological limitations and patient needs, adapting workflows, improvising during system failures, and providing reassurance to patients. These actions demonstrate that the effectiveness of digital health systems depends not only on technical functionality but also on the capacity of human actors to respond to system challenges.

This finding aligns with prior research showing that organizational resilience in digital health systems is often sustained through staff adaptability, teamwork, and problem-solving in the face of infrastructural constraints [13,14]. In this study, human intervention emerges as a key factor in maintaining service continuity, particularly in contexts where technical systems are unreliable.

6.4. Sociotechnical interdependence and implications for digital health systems

Taken together, the findings demonstrate that digital health systems operate as interdependent sociotechnical configurations, where patient experiences are shaped by the interaction between infrastructure, technology, organizational processes, and human actors. The effectiveness of these systems is therefore contingent upon achieving balance and alignment across these components.

In the context of Northern Ghana, this alignment remains uneven. While digital systems have improved efficiency and coordination, persistent infrastructural instability continues to limit their full potential. This highlights the need to move beyond technology-centered approaches to digital health implementation and adopt holistic strategies that address both technical and social dimensions.

7. Practical and policy implications

From a policy perspective, the findings emphasize the need for a holistic digital health strategy that integrates infrastructural investment with human-centered design. Policymakers should recognize that technological adoption does not automatically translate to patient satisfaction; rather, it is mediated by contextual factors such as language, literacy, and institutional responsiveness. The observed transition from LHIMS to newer systems highlights the necessity of gradual, consultative migration strategies that consider patient and staff experiences to maintain continuity and trust.

Practically, the study underscores that sustainable digital health transformation in Ghana, and similar LMIC contexts, depends on building sociotechnical resilience: reliable infrastructure, continuous technical support, staff training, and culturally attuned patient engagement. When these components align, digital systems can evolve from fragile tools into enablers of equitable, patient-centered healthcare.

8. Conclusion

This study examined patients' experiences of digital health systems across three levels of healthcare delivery in Northern Ghana. Guided by Sociotechnical Systems Theory (STS), the findings show that patient experiences are shaped by the interaction between technological systems, infrastructural conditions, and human actors. Digitalization has improved service organization, record management, and coordination, enhancing efficiency and continuity of care. However, these benefits are uneven and highly dependent on infrastructural stability, particularly electricity and network reliability, which remain significant constraints in resource-limited settings. The findings further demonstrate that digital health systems function as fragile sociotechnical configurations, where system effectiveness depends not only on technology but also on the capacity of healthcare workers to mediate and sustain service delivery during disruptions.

Importantly, the study underscores the need for continuous assessment of digital health systems, as patient experiences and system performance evolve over time. Such evaluations should extend beyond settings undergoing initial digital transformation to include digitally mature environments, where ongoing changes in technology and service delivery require regular monitoring and adaptation. Furthermore, while this study focused on patients' perspectives, it highlights the importance of incorporating healthcare professionals' experiences to achieve a more comprehensive understanding of digital health implementation. Future research should therefore adopt a more inclusive approach that integrates both patient and provider perspectives to better capture the full scope of sociotechnical dynamics in healthcare systems.

CRediT authorship contribution statement

Mohammed Saani Abdulai: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Eliasu Mumuni:** Writing – review & editing, Supervision, Methodology. **Muhammed Abdulai:** Writing – review & editing, Supervision, Methodology.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.dialog.2026.100319>.

Data availability

The data supporting the findings of the present study are available

upon reasonable request from the corresponding author. The complete dataset cannot be disclosed publicly due to ethical constraints and participant confidentiality.

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